

FUNCTIONAL GAME REFERENCE · BASED ON ACTUAL IMPLEMENTED CODE · V0.1.11

Patient Angler: Idle Fishing

Build 17 · July 2026 · Android (Google Play) · Capacitor 7 · HTML/JS · no framework

· Recapsa 0Ü / Henly Games

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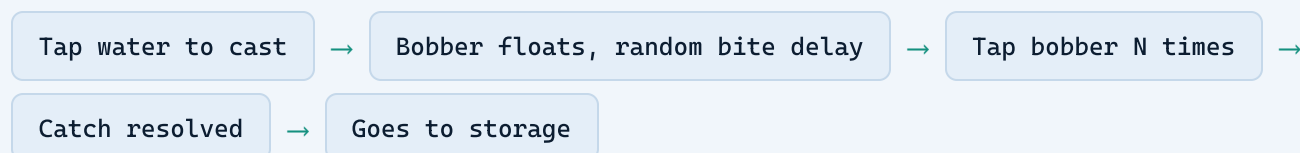
What the game is

An idle/incremental fishing game for Android. The player starts at a small pond with a basic rod and manually fishes one cast at a time. Over time they buy automation (nets, fishermen, boats, fleets) for each zone, and prestige with Black Pearls to unlock permanent bonuses. The game runs entirely in the browser via Capacitor — all logic is client-side JavaScript, save state is localStorage.

The core loop is: catch fish → sell fish for coins → spend coins on upgrades. There is no dynamic market demand and no fish spoilage — fish sit in storage indefinitely until sold at a fixed price.

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Fishing mechanic



Tap count (N) is set by the equipped rod: Basic Rod = 20 taps, down to Abyss Rod = 4 taps. The tap count is the only visible progression stat on rods — there are no hidden percentages. Players immediately feel the upgrade.

Multi-catch is possible with the Electronic Bobber and the Pearl Shop multi-catch upgrade. Each level of multi-catch (bought with Black Pearls, 5 pearls/level) adds one extra simultaneous catch per cast.

Bite speed is affected by: Rapid Waters buff ($\times 1.5$), Fishing Frenzy buff (+25%), Remove Ads permanent bonus (+25%), Pearl Shop speed upgrade (up to $3\times$). These stack multiplicatively.

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Zones

Six zones unlock sequentially. Each requires owning a specific rod AND a transport vehicle. Both cost coins from the shop. Once unlocked, the zone stays unlocked across prestiges — only coins, automation, and transport reset on prestige.

Zone unlock costs scale with prestige count: $\text{baseCost} \times (1 + \text{prestigeCount})$. First prestige makes everything 2× more expensive, making each subsequent run harder to advance despite the sell bonus.

ZONE	REQUIRES	BASE UNLOCK COST (ROD + TRANSPORT)
Pond	Basic Rod (free)	Free
River	River Rod + Waders	~2,300c
Lake	Lake Rod + Rowing Boat	~6,500c
Bay	Bay Rod + Speedboat	~15,000c
Sea	Sea Rod + Fishing Vessel	~45,000,000c
Ocean	Ocean Rod + Research Vessel	~450,000,000c

Each zone has 30 catchable items (10 fish, 10 plants, 10 trash). Each successive zone has a lower trash/plant probability and higher rare/epic probability in its loot table. Ocean is the only zone with a Legendary tier in its base loot table.

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Loot system

Every catch rolls two things independently: **loot type** (from the zone's weighted table) and **fish size** (from a shared size table).

Loot table (Pond example)

Trash 30% · Plants 20% · Common fish 35% · Uncommon 10% · Rare 4% · Epic 1%. Trash auto-sells for 1c. Plants are discarded (0c). Fish go to storage.

Size table (all zones)

Tiny 20% (×0.5) · Small 30% (×0.8) · Medium 30% (×1.0) · Large 15% (×1.5) · Trophy 5% (×3.0). Trophy also triggers a special catch fanfare and records the fish weight in Fishdex.

Sell price formula

Premium Bait effect

$\text{baseValue} \times \text{sizeMult} \times \text{rodSellBonus} \times \text{pearlBonus}$. Rod sell bonus: +15% per Basic Rod tier. Pearl bonus: $\sqrt{\text{totalPearls}} \times 0.5$, capped at +300% (4× max). No demand multiplier exists.

Doubles the weight of Uncommon/Rare/Epic/Legendary entries in the loot table for 30 minutes. Common probability drops proportionally. Activated with Diamonds (2 per use). Does not guarantee rare catches.

RARITY	BASE VALUE RANGE (POND)	BASE VALUE RANGE (OCEAN)
Common	2–4c	65–80c
Uncommon	9–12c	220c
Rare	15–18c	420–480c
Epic	55c	550c
Legendary	— (not in Pond)	560–1800c

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Market & storage

Fish go into a grouped storage pile (`fishPile` keyed by `fishId|size`) and stay there indefinitely — there is no spoilage or expiry timer. The Market screen shows all stored fish with their current sell value and a "Sell All" button.

Storage tier determines the maximum number of fish that can be stored. When full, automation pauses — no new catches are added until space is freed. 7 tiers: Bucket (5 slots) → Ice Box → Cooler Box → Portable Fridge → Chest Freezer → Walk-in Freezer → Harbor Cold Storage (10,000 slots).

Auto-Sell (permanent, bought with Diamonds): when enabled, every caught fish sells immediately at catch time. Trophy fish are exempt — they always go to storage. Auto-Sell can also be activated temporarily via a special event.

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Automation

Automation is bought per zone. Each unit is stackable — you can own multiple of the same type in the same zone. Cost per additional unit escalates: $\text{baseCost} \times 1.15^{\text{ownedCount}}$. All automation across all

zones runs simultaneously.

Automation catches Common through Epic fish only. Legendary fish require active manual fishing. Offline progress is calculated on app launch (capped at 12 hours) and added to storage immediately.

TIER	ITEMS	FIRST COST	CATCH RATE
Nets	Fishing Net / Reinforced Net / River Net	100c	1/60s → 1/30s
Fishermen	Local / Skilled / Veteran Fisher	3,000c	1/30s → 1/8s
Boats	Row / Motor / Fishing Boat	30,000c	1/8s → 1/1.5s
Fleets	Small / Large / Deep Sea Fleet	5,000,000c	2/s → 10/s

The HUD shows a live fish/s counter summing all active automation rates.

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Rods & shop

Each rod reduces the tap count required to land a fish. Rods also have a tiered upgrade system — each tier gives a permanent +15% sell bonus (Basic Rod) or other bonuses. Tier upgrades are paid separately from the initial rod purchase.

ROD	TAPS TO CATCH	INITIAL COST	TIER BONUS
Basic Rod	20	Free	+15% sell price / tier
River Rod	18	1,500c	+10% bite speed / tier
Lake Rod	16	4,000c	+15% automation rate / tier
Bay Rod	14	9,000c	+20% storage capacity / tier
Sea Rod	12	25,000,000c	+25% rare catch chance / tier
Ocean Rod	10	250,000,000c	+30% legendary chance / tier
Carbon Rod	8	200,000,000c	+40% all bonuses / tier
Mythic Rod	6	500,000,000c	+45% all bonuses / tier
Abyss Rod	4	1,500,000,000c	+50% abyss fish sell value / tier

Bobbers: Basic (free) · Sensitive (+1s tap window) · Heavy (fewer false bobs) · Electronic (haptic on bite, enables multi-catch).

Seagull Bait: Temporary coin-per-catch boost. Tiered, costs coins. Price escalates steeply with tier:
`10,000c × 5^tier`.

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Time of day system

The in-game clock runs 24× faster than real time: 1 real hour = 1 in-game day. The clock is driven by the real device clock (real minutes + seconds map to in-game hours + minutes). Always shown in the HUD.

Certain fish have a `timeWindow` property — they only appear in the loot pool during specific in-game hours. If no fish of the required rarity are available in-window, the game falls back to the full pool.

FISH	ZONE	WINDOW (IN-GAME)	RARITY
Morning Perch	Pond	07:00–10:00	Uncommon
Afternoon Roach	Pond	12:00–15:00	Common
Dawn Trout	River	04:00–07:00	Rare
Midnight Eel	River	00:00–03:00	Epic
Evening Catfish	Lake	18:00–22:00	Uncommon
Night Pike	Lake	22:00–02:00	Rare
Pre-dawn Zander	Lake	03:00–05:00	Rare
Giant Squid	Ocean	00:00–02:00	Legendary
Coelacanth	Ocean	01:00–03:00	Legendary

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Special events & buff system

Special events appear on the right edge of the fishing screen as a pulsing side icon. They fire every 15–30 minutes randomly. The player taps to open a popup and can claim or ignore. Unclaimed events expire after 90 seconds, after which the next event is scheduled.

When a buff is active, a small icon with a countdown timer appears on the left edge of the screen (buff bar). Multiple buffs can be active simultaneously.

EVENT	EFFECT	DURATION
Fishing Frenzy	+25% fishing (bite) speed	30 min
Lucky Hook	All manual catches are Rare or better	3 min
Rapid Waters	Fishing speed ×1.5	20 min
Care Package	Instant coins (~30 min of current automation income, max 12,000c)	Instant
Lost Treasure	+3 Diamonds instantly	Instant

Lucky Hook detail: when active, the loot table is filtered to only rare/epic/legendary weights for the full 3-minute duration. All manual casts during this window benefit. Timer shows in the HUD buff bar.

Remove Ads users get Lucky Hook auto-triggered every 10 minutes and a permanent +25% bite speed (equivalent to Fishing Frenzy always active).

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Prestige – Black Pearls

Prestige is available once the player has at least 15,000 coins. The player converts their current coin pile into Black Pearls, then starts over in a harder (more expensive) but faster (sell bonus) run.

What resets
Coins · All automation (nets/fishermen/boats/fleets) · Storage tier · Transport vehicles · Zone access (back to Pond only)

What is kept
Fishdex entries · Rod tier levels · Bobber tier levels · Diamonds · Black Pearls · Prestige count · Pearl Shop upgrades · Quests & achievements

Pearl reward formula
`floor(sqrt(coins / 2000))` — examples:
15,000c → 2 pearls, 50,000c → 5 pearls, 200,000c → 10 pearls, 1,000,000c → 22 pearls. Incentivises saving before prestige.

Passive sell bonus
All-time pearls give `+50% × sqrt(totalPearls)` to every fish sold. Capped at +300% (4× sell value). Accumulates forever — pearls never reset.

PEARL SHOP UPGRADE	EFFECT PER LEVEL	PEARL COST / LEVEL	CAP
Shop Discount	–10% all shop costs	2	–50%

PEARL SHOP UPGRADE	EFFECT PER LEVEL	PEARL COST / LEVEL	CAP
Fishing Speed	+25% bite speed	3	3× speed
Expanded Holds	+50% storage capacity	2	No cap listed
Multi-Catch	+1 simultaneous catch per cast per 5 levels	5	No cap listed

Cost scaling: Every prestige multiplies all zone unlock costs by $(1 + \text{prestigeCount})$. After 3 prestiges, River costs 4× more than the first run. This prevents trivially blazing through zones with accumulated sell bonuses.

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Competitions

The player manually joins a competition from the Competition screen. Duration: 5 minutes (prototype value — intended to be 60 minutes in production). The competition takes place in the player's current zone.

Opponents are 9 AI bots with names (FisherKing, Angler99, etc.). Every 10 seconds each bot has a 65% chance to make a catch attempt. Trophy rate during competition is doubled. The player fishes normally — all manual catches are tracked, and the highest single-catch value is the competition entry.

After 5 minutes the leaderboard finalises. Coin rewards:

RANK	COIN REWARD
1st	500c + Hall of Fame entry
2nd	300c
3rd	200c
4th–5th	100c
6th–10th	75c

Competitions are entirely local/offline — no server leaderboard. The bots are simulated client-side. Winning 1st saves the win to the local Hall of Fame for that zone.

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Fishdex

Every catchable item has a Fishdex entry — 180 total across 6 zones (30 per zone: 10 fish + 10 plants + 10 trash). Uncaught entries show as a grey locked silhouette. First catch unlocks the entry permanently (kept across prestige).

The Fishdex screen has scrollable zone tabs. Each zone shows a 4-column grid with fish / plants / trash sections and a progress counter. Trophy catches record their weight as a personal best on the entry.

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Quests & achievements

Three daily quests are drawn randomly at midnight from a pool of 8. One weekly quest resets on Monday. Nine permanent one-time achievements. All rewards are coins — no functional gear is rewarded (keeps economy clean). A red pulsing badge appears on the Quests nav button when rewards are unclaimed.

Progress is tracked via event hooks fired on every catch (`onCatchEvent`), every sell (`onSellEvent`), and storage-full events. Stats tracked include total fish caught, total trash caught, total epic catches, streak days, hourly catch count, surge sales.

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Diamond store

Diamonds are a secondary currency. Earned via Lost Treasure events (+3 per event), and purchased via IAP (not yet live). Spent in the Diamond Store on permanent features and consumable boosts.

Auto-Sell (permanent)

Catches auto-sell immediately at base price.
Trophy fish still go to storage. One-time purchase with Diamonds.

Temp Auto-Sell

Auto-sell active for a limited duration. Cheaper than permanent version.

Premium Bait

Remove Ads (IAP)

2 Diamonds per activation. Doubles non-common loot weights for 30 minutes. Shows as buff icon in left bar.

Removes ads, permanent +25% speed, Lucky Hook every 10 min. Stub — payment not yet implemented.

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Tech stack

- **Frontend:** Single HTML/JS/CSS app — `index.html` , `game.js` , `styles.css` . No framework, no build step.
- **Native:** Capacitor 7 on Android. Package: `com.henlygames.patientangler` .
- **Save:** `localStorage` key `patientAngler_v1` . No server save yet.
- **Backend:** Node.js/Express on Railway. Currently used only for version checks (soft update banner / forced update overlay). Competition and Hall of Fame are local-only.
- **AdMob:** `@capacitor-community/admob` v6.2.0. Production approved July 2026. Rewarded ads trigger via special events.
- **IAP:** `com.android.billingclient:billing` integrated but purchase flow is a stub — no actual transactions yet.
- **Auth:** Firebase plugin present, Google Sign-In not yet wired.
- **Orientation:** Locked to landscape via `sensorLandscape` in AndroidManifest.
- **Signing:** Shared tile-royale keystore.

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What is not yet built

PLACEHOLDER Abyss gameplay

Navigation screen with a landscape image exists. No Abyss fish, no layer system, no Diamond rewards from Abyss. Only the visual shell.

PLACEHOLDER Hall of Fame

The Hall of Fame screen shows seeded fake entries (MasterAngler, TroutHunter etc.) and records real player 1st-place wins locally. No server leaderboard.

STUB IAP purchases

STUB Google Sign-In / cloud save

Billing library is integrated. Buttons exist in the Diamond Store. No actual Play Store products registered or receipt validation — purchases simulate locally.

Firebase plugin installed. No login UI wired. All saves are local only — switching devices loses progress.

NOT BUILT Push notifications

NotificationManager stub class exists. FCM and Firebase not connected. No notifications are sent.

NOT BUILT Season system

Seasonal fish entries exist in the fish DB with time windows. No season rotation, no seasonal events, no seasonal quests activated.

Dev Log -- v0.1.13 (2026-07-09)

All changes since v0.1.12

BLACK PEARL SYSTEM (reworked)

Passive Sell Bonus

Formula: $\text{sellMultiplier} = 1 + (\text{blackPearls} \times 0.01)$

Each unspent Black Pearl adds +1% to all fish sell value.

No cap -- scales linearly with pearl count.

Spending pearls in the Pearl Shop reduces the passive bonus.

Prestige Threshold (dynamic)

Formula: $\text{threshold} = \text{floor}(15,000 \times 1.20^{\text{prestigeCount}})$

Prestige 0: 15,000c Prestige 5: ~37,324c

Prestige 10: ~92,876c Prestige 20: ~574,856c

Each subsequent prestige requires more coins to unlock.

PEARL SHOP (reworked)

Upgrade Table

Cost formula: $\text{ceil}(\text{baseCost} \times \text{growthRate}^{\text{currentLevel}})$

Market Discount	base:2 x1.60	max:5	-10% shop costs / level
Empire Boost	base:2 x1.45	max:8	+25% automation speed / level
Expanded Holds	base:2 x1.35	no cap	+50% storage capacity / level
Hauling Nets	base:5 x1.40	no cap	+1 extra auto catch / 5 levels
Lucky Waters	base:3 x1.45	max:10	+5% Rare/Epic catch chance / level
Master Angler	base:3 x1.35	max:8	-1 tap required / 2 levels (min 4)
Treasure Hunter	base:4 x1.45	max:10	+5% Lost Treasure event chance / level
Offline Expert	base:3 x1.40	max:12	+10% offline earnings / level
Competition Spirit	base:3 x1.35	max:10	+10% competition coin rewards / level
Fish Whisperer	base:4 x1.50	max:10	+3% Trophy fish chance / level

TARGETED LURE (new system)

Overview

Permanent coin upgrade, 5 levels. Purchased from Bait Shop tab.
Each level adds 1 active target slot.
Targets are set from Fishdex -- tap "Target" on any caught eligible item.
Eligible: common / uncommon / rare / epic fish, plants, trash.
Not eligible: legendary fish, special/manual-only items.

Costs & Effect

Level 1 -- 25,000c -> 1 target slot
Level 2 -- 250,000c -> 2 target slots
Level 3 -- 2,500,000c -> 3 target slots
Level 4 -- 25M coins -> 4 target slots
Level 5 -- 250M coins -> 5 target slots

Targeted items get x3 loot weight in automation rollCatch().
Does NOT affect manual (active) fishing -- automation only.

SEAGULL REWARD (rebalanced)

Formula

Reward = $\max(10, \text{round}(\text{getEstimatedHourlyIncome()} \times 0.01 \times \text{baitMult}))$

Estimated hourly income = base automation income only.

-- No Pearl bonuses, no rod sell bonus, no multipliers.

Bait multiplier: +10% per Seagull Bait consumed (linear, no cap).

$\text{baitMult} = 1 + (\text{seagullBaitCount} \times 0.10)$

Scales throughout the entire game with player progression.

PREMIUM BAIT

Updated Cost

Cost: 5 Diamonds (increased from 2)

Effect: unchanged -- doubles non-common loot weights for 30 min.

Does not guarantee rare catches; improves expected value.

SERVER-SIDE REDEEM CODES

Architecture

Codes stored on Railway server -- not visible in client source code.

Endpoint: POST /pa/redeem { uid, code }

Each code can be redeemed once per Google account (uid).

Requires Google Sign-In to redeem.

Table: pa_codes_used (code TEXT, uid TEXT, PRIMARY KEY(code, uid))

Active codes: REVIEW, LAUNCH, PEARLS5

Reward types: coins, diamonds, autoIncome (1h estimated income)

FISHDEX MASTERY (new system)

Mastery Tiers (per item)

Bronze -- 100 catches -> 1 mastery point
Silver -- 1,000 catches -> 2 mastery points
Gold -- 10,000 catches -> 3 mastery points
Platinum -- 100,000 catches-> 4 mastery points
Diamond -- 1,000,000 catches -> 5 mastery points

Eligible: all fish except legendary/special, all plants, all trash.
All catches count (manual + automation + offline).
Counts persist across prestige (masteryData kept).

Zone Mastery Points

Zone points = sum of all item tier points in that zone.
Max 150 points per zone (30 items x 5 pts).
Bonus thresholds: 25 / 50 / 75 / 100 / 125 / 150 points.

Zone Bonuses (unlocked at each threshold)

Pond: +5% fish sell, +10% storage, +3% rare catch,
+8% auto speed, +10% quest reward, +10% offline
River: +5% fish sell, +8% auto speed, +3% rare catch,
+10% offline, +2% trophy chance, +10% comp reward
Lake: +5% fish sell, +10% storage, +3% rare catch,
+8% auto speed, +10% offline, +2% trophy chance
Bay: +5% fish sell, +3% rare catch, +2% trophy chance,
+10% offline, +10% quest reward, +8% auto speed
Sea: +5% fish sell, +10% storage, +3% rare catch,
+2% trophy chance, +8% auto speed, +10% comp reward
Ocean: +5% fish sell, +3% rare catch, +2% trophy chance,
+10% storage, +10% offline, +10% comp reward

Fishdex UI

Zone mastery panel at top: points bar + bonus tags (locked/unlocked).
Each item cell shows: tier medal (B/S/G/P/D), catch count, progress bar.
Diamond tier cells have animated cyan glow border.

COMPETITION REWARDS (rebalanced)

New Formula

1st Place: $500 + \text{round}(\text{estimatedHourlyIncome} \times 0.05)$
2nd Place: $250 + \text{round}(\text{estimatedHourlyIncome} \times 0.03)$
3rd Place: $100 + \text{round}(\text{estimatedHourlyIncome} \times 0.01)$
4th+ Place: 0

Estimated hourly income = base economy only.

-- No Pearl/Mastery bonuses, no rod sell multiplier.

Pearl Competition Spirit and Mastery compReward bonuses
apply on top as multipliers after the base reward is calculated.

Competition gameplay, duration, and AI unchanged.

ROD TIER EFFECTS (rebalanced)

New Values

Basic Rod	+5% fish sell price / tier	(was +15%)
River Rod	+10% Net automation speed / tier	(was +20%)
Lake Rod	+10% Fisherman speed / tier	(was +20%)
Bay Rod	+12% storage capacity / tier	(was +20%)
Sea Rod	+10% Boat speed / tier	(was +20%)
Ocean Rod	+10% Fleet speed / tier	(was +20%)
Carbon Rod	+3% Legendary catch chance / tier	(was +10%) (relative weight multiplier, not flat %)
Mythic Rod	+8% Diamond earnings / tier	(was +50%)
Abyss Rod	+8% Abyss fish sell value / tier	(was +50%) (Abyss zone fish only -- unchanged scope)

Rod unlock costs and click counts unchanged.

Existing tier levels carry over -- only percentages changed.

ADMOB GDPR FIX

UMP Consent Flow

EU/EEA players (e.g. Greece, Estonia) were not receiving ads.

Root cause: AdMob requires UMP consent before initialization.

Fix: `requestConsentInfo()` -> `loadAndShowConsentFormIfRequired()`

runs before `AM.initialize()` on every app start.

If consent SDK unavailable, falls back gracefully and continues.